The Integrating Factor as Operator Conjugation

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Introduction

Consider the differential equation $y'' - 5y' + 6y = 0$. To solve, the traditional approach is to guess solutions of the form $y = e^{rt}$. This leads us to the characteristic polynomial

$$r^2 - 5r + 6 = 0,$$

which has roots $r = 2$ and $r = 3$. So, the general solution is $y = c_1 e^{2t} + c_2 e^{3t}$.

Introduction

We can verify that this works, but we're left with some questions:

- Why do the exponentials work?
- Why do we multiply by t whenever there is a repeated root?
- How do we know these are all the solutions?

To answer these questions, we revisit a familiar technique.

The Integrating Factor

Consider the first-order equation $y' - \lambda y = 0$. Recall that the integrating factor method says to multiply by a function $\mu(t)$ to obtain

$$(\mu y)' = 0.$$

Key observation: The left-hand side is transformed into an exact derivative, which is directly integrable.

The Operator Perspective

To see why this works, we shift perspective. Instead of the solving the equation $y' - \lambda y = 0$, we study the operator $D - \lambda I$ acting on smooth functions.

We define $\ker(D - \lambda I) := \{y : (D - \lambda I)y = 0\}$. This means that the kernel describes precisely the set of solutions to the differential equation.

The Core Discovery

Lemma (Conjugation Identity)

For any $\lambda \in \mathbb{C}$, if M_λ denotes multiplication by $e^{-\lambda t}$, then $M_\lambda^{-1} D M_\lambda = D - \lambda I$.

Proof.

$M_\lambda^{-1} D M_\lambda y = e^{\lambda t}(y \cdot e^{-\lambda t})' = e^{\lambda t}(y' e^{-\lambda t} - \lambda y e^{-\lambda t}) = y' - \lambda y.$ $\square$

This identity says that multiplying by $e^{-\lambda t}$ *conjugates* the operator $D - \lambda I$ into the simple operator D.

Core discovery: The integrating factor is operator conjugation in disguise!

Higher-Order Equations

Iterating the conjugation identity m times results in

$$M_\lambda^{-1} D^m M_\lambda = (D - \lambda I)^m.$$

This lets us *derive* the solutions to higher-order equations of the form $(D - \lambda I)^m y = 0$:

Corollary

$\ker(D - \lambda I)^m = M_\lambda^{-1} \ker D^m = \operatorname{span}\{e^{\lambda t}, te^{\lambda t}, \ldots, t^{m-1} e^{\lambda t}\}$. Moreover, $\dim \ker(D - \lambda I)^m = m$.

Main Result

Now we consider a more general problem: solving any constant-coefficient equation, or equivalently, $p(D)y = 0$ where $p(z)$ is any polynomial.

Theorem

Suppose $p(z) = \prod_{j=1}^{k}(z - \lambda_j)^{m_j}$, where $\lambda_1, \ldots, \lambda_k$ are distinct complex numbers and $m_1, \ldots, m_k$ are positive integers. Then $\ker p(D) = \bigoplus_{j=1}^{k} \ker(D - \lambda_j I)^{m_j}$.

Conjugation Example

Consider the differential equation $y''' - 5y'' + 3y' + 9y = 0$. First, we factor the operator into $(D - 3I)^2(D + I)y = 0$.

Conjugation Example

The equation $(D-3I)^2(D+I)y=0$ is equivalent to the system $(D-3I)^2u=0$ and $(D+I)y=u$. Using the conjugation identity, rewrite the system as

$$M_3^{-1}D^2M_3u=0 \quad \text{and} \quad M_{-1}^{-1}DM_{-1}y=u.$$

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$$M_{-1}^{-1}DM_{-1}y=M_{-3}p_2 \implies y=M_1JM_{-4}p_2,$$

where J represents integration. Integration by parts gives us (with new constants C_0, C_1, C_2)

$$y=C_0te^{3t}+C_1e^{3t}+C_2e^{-t}.$$

Extensions and Limitations

- The operator conjugation framework also applies to special variable-coefficient cases where commutativity holds.
- In general, equations with variable coefficients cannot be broken apart into commuting operators. For example,

$$(D - tI)(D + tI)y \neq (D + tI)(D - tI)y.$$

- This highlights why constant coefficient equations are special: their operators *always* commute!

Conclusion

Recognizing the integrating factor as operator conjugation unlocks the solution structure of constant-coefficient differential equations:

- The exponentials ($e^{\lambda t}$) emerge from conjugation via M_λ.
- The polynomial factors emerge from $\ker D^m = \{1, t, \dots, t^{m-1}\}$.
- The direct sum decomposition emerges from factoring $p(D) = (D - \lambda_1 I)^{m_1} \dots (D - \lambda_k I)^{m_k}$ and operator commutativity.
- The identity $(D - \lambda I)^m = M_\lambda^{-1} D^m M_\lambda$ reveals that we have *all* the possible solutions.

Crucially, we did not guess these solution forms. Instead, we have **derived** them by applying this framework.

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Thank you for your time!